EXPERIMENT -1 :CODE

```
const int GAS_PIN = A0;
const int PIR_PIN = 2;
const int TRIG_PIN = 9;
const int ECHO_PIN = 8;
const int LED_PIN = 11;
const int BUZZER_PIN = 10;

const int GAS_LIMIT = 400;
const float DISTANCE_LIMIT = 50.0;

void setup() {
  pinMode(GAS_PIN, INPUT);
  pinMode(PIR_PIN, INPUT);
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  Serial.begin(9600);
  Serial.println("GAS AND MOTION SECURITY SYSTEM");
  Serial.println("==============================");
}

void loop() {
  // Gas sensor reading
  int gasValue = analogRead(GAS_PIN);
```

```
// Convert reading to a relative percentage, not actual gas concentration
int gasLevel = map(gasValue, 0, 1023, 0, 100);

// Motion detection
bool motionDetected = digitalRead(PIR_PIN) == HIGH;

// Ultrasonic distance measurement
digitalWrite(TRIG_PIN, LOW);
delayMicroseconds(2);
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);

unsigned long duration = pulseIn(ECHO_PIN, HIGH, 30000UL);
bool distanceValid = duration > 0;
float distance = duration * 0.0343 / 2.0;

// Determine system status
bool gasWarning = gasValue >= GAS_LIMIT;
bool nearbyObject = distanceValid && distance <= DISTANCE_LIMIT;
bool systemSafe = !gasWarning && !motionDetected && !nearbyObject;

// Alarm outputs
digitalWrite(LED_PIN, systemSafe ? LOW : HIGH);
digitalWrite(BUZZER_PIN, systemSafe ? LOW : HIGH);

// Print required information
Serial.print("Gas Level: ");
Serial.print(gasLevel);
Serial.print("% (Raw: ");
Serial.print(gasValue);
Serial.println(")");
```

```
  Serial.print("Motion: ");
  Serial.println(motionDetected ? "DETECTED" : "NOT DETECTED");

  Serial.print("Distance: ");
  if (distanceValid) {
    Serial.print(distance, 1);
    Serial.println(" cm");
  } else {
    Serial.println("NO ECHO");
  }

  Serial.print("System Status: ");
  Serial.println(systemSafe ? "SAFE" : "WARNING");

  Serial.println("------------------------------");
  delay(1000);
}
```